

1 Sharing sunlight between trackers and crops: dynamic
2 light-sharing control for agrivoltaics under a
3 food-security floor on the world's staple croplands

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6 **Abstract**

Agrivoltaic systems grow crops under photovoltaic (PV) modules, but the electricity they can produce without compromising food production is usually set by a fixed array geometry. Single-axis trackers offer another lever: the daily rotation can be shifted from maximum electricity towards letting light through, a choice we call light sharing. We quantify how much this lever is worth under a food-security floor (90% of the open-field yield) and how robust the answer is. We built a differentiable simulator that couples solar geometry, trackers with backtracking, PV output, crop light and microclimate, a soil-water balance and the SIMPLE crop model with a flowering-stage shade penalty. The PV part agrees with PVGIS at 29 sites ($r = 0.97$, mean bias -4.9%). The crop response to shade was compared with a meta-analysis and with field trials of rice, wheat, potato and maize. The simulated response is close to proportional and is conservative against the trials, which makes the reported gains lower bounds. All methods were compared in a deployable protocol: the array density of every site was chosen from its historical seasons with one rule, and performance was measured on geographically held-out sites and years, for 400 crop-site pairs representing the harvested area of wheat, rice, maize, soybean and potato, in baseline and $+1.5$, $+2$ and $+3$ °C climates. At equal mean yield retention (about 0.905), simple light-sharing rules allowed arrays about three times as dense as static arrays and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24 and electricity from 217 to 387–465 MWh ha⁻¹ yr⁻¹. A rule that follows crop phenology, which needs no weather forecast, was the best hand-written rule. It remained best, and its gain over static arrays rose from 0.17 to 0.19 and 0.30 LER when the shade response was calibrated or fitted to the field trials. A neural controller

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Preprint submitted to Artificial Intelligence in Agriculture
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trained through the simulator added a smaller gain (0.03–0.05 LER, 84–86% of held-out sites), consistently over seeds, splits, canopy-cooling assumptions and climates, but it learned a smooth function of phenology rather than a reaction to heat or drought. The floor was met in 70% of individual seasons, and production-weighted retention was below 0.9 for soybean and maize, so per-site enforcement needs region- and crop-specific design. Equipping 5% of the world’s staple cropland at the floor would yield about 14 PWh yr^{−1} but would need about 10 TWp of modules, 5.7 times the installed global PV capacity.

7 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,
8 Crop modelling, Differentiable simulation

9 1. Introduction

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017), and a new generation of climate and
17 crop models shows these impacts emerging earlier than previously expected
18 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
19 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
20 rural electricity demand, so converting farmland into solar farms is tempting
21 and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-

date adaptation measure against heat and drought stress, in addition to a mitigation measure through renewable electricity. The same shade, however, also reduces the light available for photosynthesis. Regulators increasingly require AV systems to retain most of the reference crop yield. The German pre-standard DIN SPEC 91434, for example, requires at least 66% of the reference yield (DIN, 2021), and other schemes and many farmers set stricter floors. How much electricity can be produced without sacrificing food security depends on the crop, the local climate, the year and the growth stage.

Most AV designs fix the geometry of the array, i.e. the ground coverage ratio (GCR, module area per ground area) and the tilt or tracking rule, from generic rules of thumb. Single-axis trackers add a degree of freedom: the modules can be rotated away from the sun to let light through when the crop needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the sun to maximise electricity. Choosing the rotation every day is a sequential decision problem. The value of light depends on the crop’s phenological stage, soil water and upcoming weather, and the effect of a decision appears in the harvest months later. Recent work in this journal has used evolutionary multi-objective optimisation to lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for crop management has mainly used reinforcement learning with black-box simulators (Gautron et al., 2022), which is sample-inefficient and hard to scale to global, multi-decadal assessments.

Two questions follow. First, how much of the electricity that an agri-voltaic system can produce under a food-security floor comes from the array density, and how much from dynamic light-sharing control of the trackers? Second, how robust is that answer to the weakest part of any such study, the crop response to shade, and does a controller learned from a simulator add anything to a well-designed hand-written rule? Answering them requires a simulator that couples the physics of light, PV, microclimate and crop growth, a protocol in which all controllers face the same design constraints, a check of the shade response against field data, and enough sites and seasons to separate real effects from noise.

This paper makes three contributions. The first is a quantification of light-sharing control under a yield floor. Light sharing is parameterised by one daily level between full backtracking and the rotation that minimises the shadow on the crop, and we compare static arrays, a seasonal schedule, a rule that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen

from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds and three spatial splits. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS, and the crop response to shade is compared with a meta-analysis (Laub et al., 2022) and with field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023).

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry about three times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The simulated shade response is conservative against published field trials. When it is calibrated to a meta-analysis or fitted to the trials, the allowed density and the gain of light sharing rise, and the ranking of methods is unchanged (Section 3.6). The gains reported here are therefore lower bounds with respect to the trials we could use.
4. A learned controller adds a smaller gain over the best rule, 0.03–0.05 LER (about 10% more electricity at equal retention), consistent over seeds, splits, canopy-cooling assumptions and climates. It is a smooth function of crop phenology, and one network serves floors between 0.80 and 0.95 (Section 3.4).
5. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr^{−1}, but only with about 10 TWp of mod-

108 ules, several times the installed PV capacity of the world (Section 3.8).

109 2. Materials and methods

110 2.1. Global cropland sites, weather and crop calendars

111 We studied five staple crops that together cover about 670 Mha of har-
112 vested area and 580 Mha of physical cropland: wheat, rice, maize, soybean
113 and potato. Harvested areas (rainfed and irrigated) were taken from the
114 SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1°
115 cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn
116 without replacement with probability proportional to harvested area. An un-
117 weighted average over the sample therefore approximates an area-weighted
118 average over global production. The 400 crop-site pairs lie in 381 distinct
119 cells in 50 countries (Fig. 2). Sowing and maturity dates came from the
120 GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat
121 we used the winter-wheat calendar unless the coldest-month mean temper-
122 ature was below -8°C , in which case the spring-wheat calendar was used.
123 Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,
124 2018): global and diffuse shortwave irradiance, maximum, minimum and
125 dew-point temperature, precipitation and wind speed at 2 m. Each site con-
126 tributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days
127 after sowing, which gives 7600 site-seasons.

128 Climate change was represented by the global warming levels +1.5, +2
129 and +3 $^{\circ}\text{C}$ relative to 2001–2020. Daily maximum, minimum and dew-point
130 temperatures were shifted uniformly, which keeps relative humidity approx-
131 imately constant and raises the vapour-pressure deficit. The atmospheric
132 CO_2 concentration was set to 390, 450, 500 and 600 ppm for the baseline and
133 the three warming levels. Precipitation and irradiance were left unchanged.
134 This delta approach isolates the thermal signal of warming; its limitations
135 are discussed in Section 4.4.

136 2.2. A differentiable agrivoltaic simulator

137 The simulator advances all site-seasons in parallel with a daily time step
138 for the crop and an hourly time step for radiation and PV (Fig. 1). All equa-
139 tions are implemented as PyTorch tensor operations (Paszke et al., 2019), so
140 that yields and electricity are differentiable with respect to the design and
141 control variables.

142 *Hourly irradiance.* Daily global and diffuse irradiance were disaggregated
 143 into hourly values with the Collares-Pereira and Rabl (1979) and Liu and
 144 Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance
 145 on the horizontal is the difference between the two. Solar position follows
 146 the standard declination and hour-angle equations.

147 *Array geometry and PV electricity.* We consider elevated horizontal single-
 148 axis trackers with north–south axes. The axis height is 1.25 module widths,
 149 e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The ro-
 150 tation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width
 151 divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this
 152 keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the
 153 rows when the modules are horizontal, which is the stow position during field
 154 operations. Let ψ be the solar zenith angle projected onto the east–west ver-
 155 tical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,
 156 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ oth-
 157 erwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on
 158 to the sun, which minimises the shadow on the crop. Given a light-sharing
 159 level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

160 Plane-of-array irradiance combines three terms: beam irradiance on the mod-
 161 ule, reduced by the fraction of the module shaded by the neighbouring row,
 162 $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected
 163 light (albedo 0.2). The cell temperature follows the NOCT model (45 °C),
 164 with an hourly air temperature interpolated sinusoidally between the daily
 165 extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with
 166 $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature
 167 losses and system losses. The installed capacity is 2.1 g MWp ha⁻¹. A con-
 168 ventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the
 169 energy reference E_{PV} .

170 *Validation of the PV model.* We compared the simulated specific yield of an
 171 isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2
 172 (European Commission, Joint Research Centre, 2022), for a horizontal north–
 173 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29
 174 returned data). Across sites, the simulated and PVGIS specific yields were
 175 correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean

absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and with the different irradiance databases (NASA POWER vs. SARAH2, ERA5 and NSRDB). Electricity estimates are therefore, if anything, conservative.

Light and microclimate at the crop. Averaged over the row pitch, the fraction of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-view factor. We pre-computed it for 25 values of g and 13 rotation angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly (differentiably). The daily ratio s of the irradiance at the crop to that of the open field defines the crop’s shading. The canopy maximum temperature is lowered by $\kappa(1 - s)$.

κ is uncertain. Field studies under PV modules report small reductions of daytime crop or canopy temperature and little change in air temperature (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds calibrated on air temperature do not transfer directly to canopy temperature (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$ for all methods (Section 3.2). Nocturnal warming under the modules is not represented. The reference evapotranspiration under the modules, ET_0 , is recomputed with the FAO-56 Penman–Monteith equation (Allen et al., 1998) from the reduced radiation and canopy temperature. Soil water conservation under the modules (Hassanpour Adeh et al., 2018; Elamri et al., 2018) therefore emerges from the model, but rain redistribution by the modules (Marrou et al., 2013a) is ignored.

Crop growth. Crop growth follows the SIMPLE model (Zhao et al., 2019) with the species parameters of its Table 1a (Table 1). Daily biomass growth is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as follows.

- f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends only on thermal time, with senescence accelerated by heat and drought stress. Shade therefore reduces the daily biomass increment but not leaf-area development.

- f_{Heat} decreases linearly between the heat threshold T_{heat} and the extreme temperature T_{ext} of the canopy maximum temperature.
- $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for drought (Woli et al., 2012) is computed from a daily soil-water balance with curve-number runoff, drainage and root water uptake. Irrigated fields have $f_{\text{Water}} = 1$.

SIMPLE has a fixed harvest index and therefore ignores the high sensitivity of grain number to shading around flowering (Fischer, 1985; Andrade et al., 1999). It also ignores the differences in shade tolerance between crop types. In the meta-analysis of Laub et al. (2022), maize and grain legumes lost yield strongly even at low shade levels, whereas C3 cereals and tuber crops initially lost less than proportionally; shading trials also show marked effects of shade on maize development (Ramos-Fuentes et al., 2023). We added a stage-dependent penalty. During a critical window ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$ is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al. (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice 0.5, for which no specific curve was available) and lowest for potato (0.2). These values are assumptions that reproduce this ranking; they are not calibrated estimates. Section 3.6 shows the resulting shade-response curves. Because RUE is linear, SIMPLE does not credit the higher light-use efficiency of shaded canopies under more diffuse light. The penalty, conversely, makes shading around flowering more damaging than proportional. The net effect is closer to the field evidence than either assumption alone.

The thermal-time requirement to maturity was calibrated for every site as the mean thermal time between the GGCM sowing and maturity dates under the 2001–2020 climate, and the canopy parameters were scaled accordingly. The cultivar is thus adapted to the present climate, and warming shortens the season. Growth stops smoothly once the thermal-time requirement is met. Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site is represented by a rainfed and an irrigated sub-field in proportion to its SPAM irrigated fraction. Each sub-field carries its own array, so both yield and electricity are area-weighted over the two sub-fields.

243 2.3. SALS: simulation-trained adaptive light sharing

244 SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 245 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 246 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 247 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 248 are the information available on day t :

- 249 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 250 terception f_{Solar} , relative soil water and the previous day’s ARID;
- 251 • a *forecast* of the day’s maximum temperature (relative to T_{heat} and
 252 T_{opt}), irradiance and ET_0 ;
- 253 • static information: the irrigation status of the sub-field, the array’s
 254 GCR g , latitude, day of season and the crop.

255 Forecasts are perturbed with random errors, independently every day: Gaus-
 256 sian with a standard deviation of 2°C for maximum temperature, and mul-
 257 tiplicative with standard deviations of 20% for irradiance and 15% for ET_0 .
 258 The physics always uses the true weather. The same forecast errors are
 259 applied to the rule-based controllers. The controller is conditioned on g .
 260 During training, each season is simulated with a GCR close to the density
 261 that is sustainable at its training site. This is the largest value on a coarse
 262 grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention
 263 of the site’s training seasons at or above ρ , plus a uniform perturbation in
 264 $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to con-
 265 trol the densities that design selection will later pick. Outside the cropping
 266 season the trackers backtrack for maximum electricity.

267 For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the
 268 yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional
 269 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

270 which combines an exact (hinge) and a quadratic penalty for the food-security
 271 floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the
 272 exact gradient of Eq. (3) is obtained by back-propagation through the 365
 273 daily steps of every season (back-propagation through time). This is the key

274 difference from reinforcement learning with black-box simulators (Gautron
275 et al., 2022), which estimates gradients from sampled returns.

276 Each minibatch contains 512 site-seasons, each simulated as a rainfed and
277 an irrigated sub-field, drawn from one of the four climates in turn. Seasons
278 with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training
279 and evaluation. The network is fine-tuned through the simulator with Adam
280 (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm
281 clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in
282 a numerically stable form, so that gradients remain finite even under extreme
283 heat.

284 *Initialisation.* Trained from random weights, the controller converged to
285 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
286 the seasonal-sharing rule, because the gradient of electricity with respect to u
287 is small and of the same sign for most days of a season. We therefore initialise
288 the network by behaviour cloning of the phenology rule of Section 2.4: binary
289 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
290 1.7 million day-states visited by the rule in the training seasons, with 1500
291 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
292 policy away from this initialisation wherever doing so improves the objective.
293 Consequently, the gain of SALS over the phenology rule measures what fine-
294 tuning through the simulator adds to that rule, not what can be learned
295 without it.

296 2.4. Deployable design selection and baselines

297 A real array has one fixed density, which must be chosen before it is built.
298 Every method, SALS included, therefore uses the same *history-based design*
299 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
300 for which the method’s mean yield retention over the site’s 2001–2014 sea-
301 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
302 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in
303 at least 80% of the historical seasons. The historical seasons of the test sites
304 are never used for training, and the weather of the evaluation years is never
305 used for design. We compare four controllers:

- 306 • *AV static*: backtracking trackers without light sharing ($u = 0$);
- 307 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
- 308 cropping season;

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
 - 2: **for** step = $1, \dots, N$ **do**
 - 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
 - 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
 - 5: **for** day $t = 1, \dots, 365$ **do**
 - 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
 - 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
 - 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
 - 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
 - 10: back-propagate through the season; Adam update of ω
-

- 309 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
310 growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress
311 forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
312 track the sun and shade the crop;
- 313 • *SALS*: the learned controller.

314 The open field and a conventional PV plant without crops ($\text{LER} = 1$ each)
315 are the references. Because the design rule is identical for all controllers, dif-
316 ferences between methods reflect the quality of control and the array density
317 that each controller can sustain under the food-security floor. We also com-
318 pare the controllers at identical GCR over the whole grid (matched-density
319 frontiers).

320 2.5. Evaluation protocol, statistics and upscaling

321 Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to train-
322 ing (70%) or test (30%), so that no test site has a training site in the same
323 block. Training uses the 2001–2014 seasons of the training sites. Design
324 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the
325 2015–2019 seasons of the test sites. The same site-seasons are analysed in all
326 four climates.

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in °C, thermal times in °C d, RUE in gMJ⁻¹; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

We report the following metrics:

- the mean electricity (MWh ha⁻¹ yr⁻¹) and the relative electricity e ;
- the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- season-level compliance (share of seasons with $r \geq \rho$) and site-level compliance (share of sites whose mean test-season retention is $\geq \rho$);
- the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

Statistical comparisons use site means as the unit of analysis. We report paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence intervals of the mean difference. SALS was trained with three random seeds on the main split, and with one seed on each of two further spatial splits.

For the global estimate, every site is evaluated in the split in which it is held out. The mean electricity and yield per hectare of each crop are scaled to the *physical* area of that crop in SPAM 2010, which avoids double counting multiple-cropped land (Waha et al., 2020). We report the installed capacity (TWp), the capacity factor and a supply curve.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out sites of the main split (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By

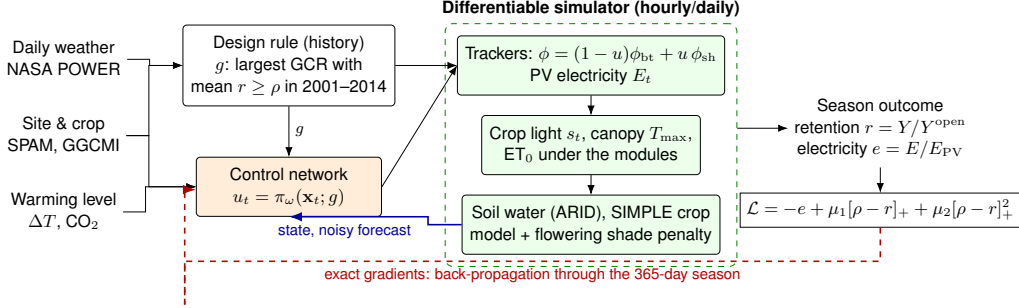


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

349 construction all methods keep the mean yield retention close to 90% (0.904–
 350 0.917), so differences in electricity are differences in the density that each
 351 controller can sustain under this floor.

352 **Static arrays are the weakest.** Without light sharing, the floor limits
 353 the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$,
 354 16% of a conventional PV plant per hectare (LER 1.07). Rotating the
 355 modules to let light through during the cropping season (seasonal shar-
 356 ing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule
 357 that shares light only while the crop is growing (phenology rule) reaches
 358 $465 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.24). An expert rule that additionally shades the
 359 crop on forecast heat or drought days (stress rule) is worse than the phenol-
 360 ogy rule (LER 1.18): its density is limited to 13% because it reduces light at
 361 times when the floor is most binding.

362 **The learned controller is best by a modest, consistent margin.**
 363 SALS sustains a density of 17% (similar to the phenology rule), produces
 364 $510 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ and reaches an LER of 1.270 at the same retention. Com-
 365 pared with the phenology rule this is +0.038 in LER (95% CI over sites
 366 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Ta-
 367 ble A.6). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35
 368 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows
 369 the same picture at matched densities: the learned policy lies on or slightly
 370 above the phenology rule over the whole density range, both lie well above

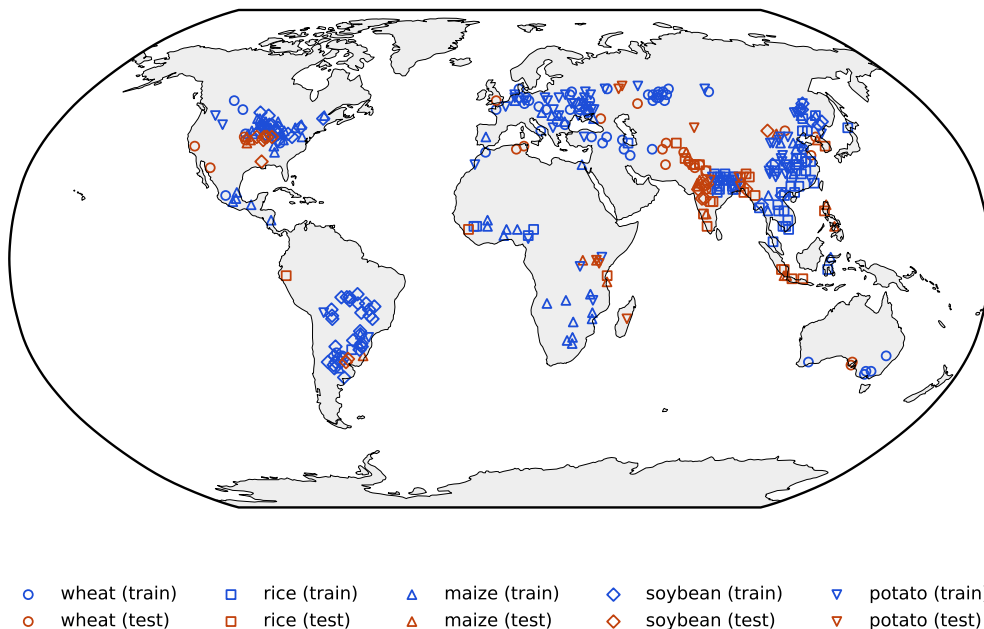


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

seasonal sharing and the stress rule, and static arrays are far below.

What the floor does and does not guarantee. Enforcing the floor on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at $+3^\circ\text{C}$. On two further spatial splits, in which different blocks of sites are

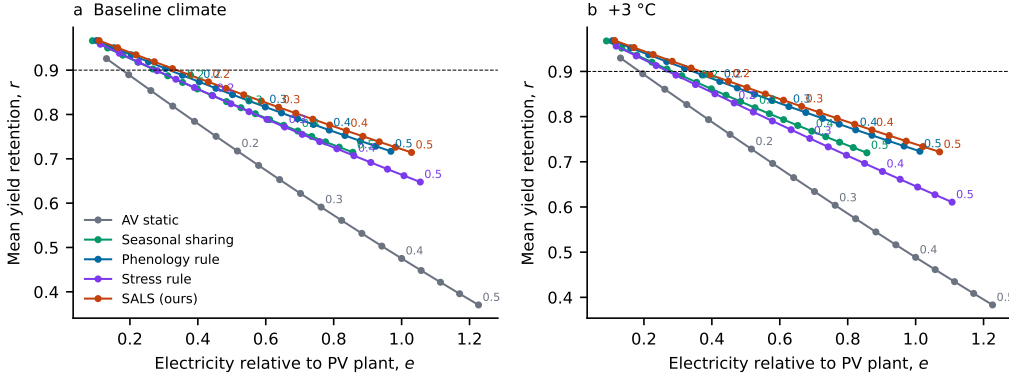


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

386 held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under
 387 the baseline climate and by 0.052 and 0.039 at +3 °C (main split: 0.034 and
 388 0.032). In every split SALS ranks first, the phenology rule second and static
 389 arrays last.

390 **Canopy cooling.** The canopy-cooling coefficient κ is the most un-
 391 certain microclimate parameter. We repeated the whole protocol, includ-
 392 ing the historical design step, with $\kappa = 0, 1$ and 2 °C. Electricity changes
 393 with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and
 394 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at
 395 0.905 by the design rule and the ranking is unchanged. SALS exceeds the
 396 phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate
 397 (0.022, 0.032 and 0.051 at +3 °C).

398 **Which ingredients matter.** Two ablations test the design of SALS
 399 at the same training budget. Masking the stress inputs (soil water, ARID,
 400 and the maximum temperature, irradiance and ET₀ forecasts) leaves the
 401 LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297
 402 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The
 403 improvement over the rules therefore does not depend on stress-aware inputs
 404 or on training in warmed climates.

405 3.3. Floor conditioning, chance-constrained training and forecast errors

406 **One network for all floors.** Regulations and farmers set different
 407 floors. The network of the main experiments was trained at $\rho = 0.9$ only.

408 We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical
 409 design step for each floor (Table 3). Its LER exceeds that of the phenology
 410 rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037,
 411 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without
 412 retraining. A network that received the floor as an additional input and was
 413 trained over all four floors did not improve on it (differences in LER ≤ 0.001).
 414 Because the design step already adapts the density to the floor, conditioning
 415 the controller on the floor is not needed.

416 **Chance-constrained training.** We also trained a network whose ob-
 417 jective penalises a batch share of seasons below the floor that exceeds 20%.
 418 It did not improve the tail. With the mean-based design step its LER was
 419 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-
 420 constrained design it reached season-level compliance of 92.6% against 92.0%
 421 and an LER of 1.198 against 1.205. The season-level guarantee is obtained
 422 by the design step (Section 3.1), not by the training objective.

423 **Forecast errors.** We raised the forecast errors seen by the controllers
 424 to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and
 425 of the phenology rule was identical at all four error levels to three decimals
 426 (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and
 427 1.416). Neither controller uses the weather forecast: both rely on the crop’s
 428 thermal time, which is observable from on-farm temperature records. The
 429 stress rule, which depends on the forecast of maximum temperature, degrades
 430 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
 431 its LER from 1.432 to 1.418 as the errors triple.

432 3.4. *What the learned policy does*

433 Averaged over the rainfed test seasons of the main split at a density of 0.3,
 434 the light-sharing level of SALS is a smooth function of relative thermal time.
 435 It is close to zero in the first 10% of the thermal time required for maturity,
 436 rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and
 437 falls through 0.58 at 80–90% to about zero at maturity. The plateau covers
 438 the window in which our crop model penalises shading through the harvest
 439 index. The ramps at both ends let the modules produce electricity when
 440 additional light is worth least to the crop. Fig. 4 shows two example seasons.
 441 The phenology rule, which is also the initialisation of the network, shares
 442 light during the whole growing period. Fine-tuning through the simulator
 443 found that sharing can be reduced at the edges of the season: SALS gains its

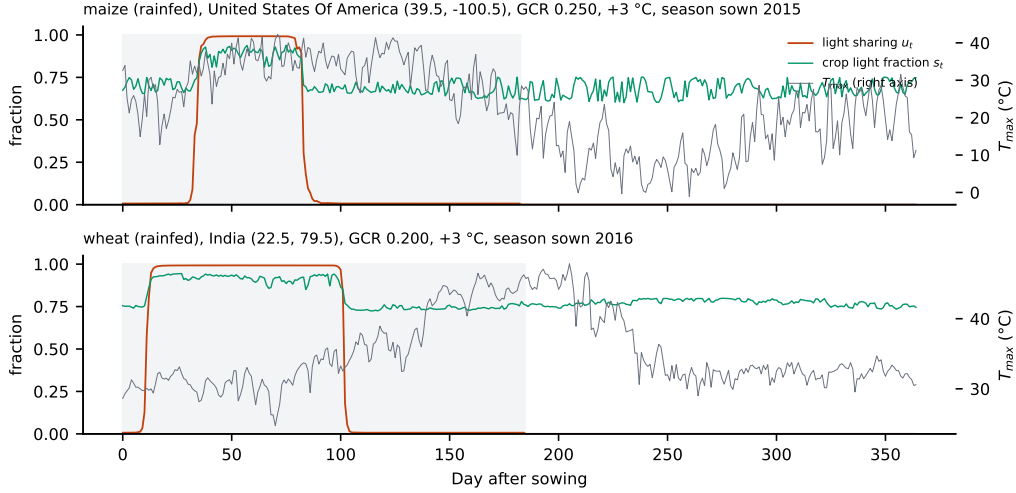


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

444 advantage by yielding light when it matters and collecting electricity when
 445 it matters less.

446 The policy also shares less light on days with a forecast heat stress: the
 447 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 ($+3^{\circ}\text{C}$)
 448 on other days, and 58–63% of sites show a lower level on heat days. This
 449 cannot be read as a stress response, however. Heat days occur late in the
 450 season, when the phenological ramp is already falling, and a policy without
 451 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).
 452 The ablation above confirms that stress information does not improve the
 453 result.

454 3.5. Crops and regions

455 The benefit of light sharing differs between crops (baseline climate, SALS).
 456 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
 457 $577 \text{ MWh ha}^{-1} \text{ yr}^{-1}$; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
 458 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
 459 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
 460 1.14–1.21 for the others).

461 The floor is not met evenly. Production-weighted retention is 0.888 for
 462 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
 463 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
 464 combine the highest density (GCR 0.41) and electricity (1090 MWh ha⁻¹ yr⁻¹)
 465 with a production-weighted retention of only 0.862. Africa and the Middle
 466 East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention
 467 of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902
 468 (67% of sites) at a density of 0.14. A regulatory floor enforced per site would
 469 therefore need a region- and crop-specific design step, not a global one.

470 3.6. Model behaviour and validation

471 **PV output.** The simulated specific yield of an isolated tracker row agrees
 472 with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29
 473 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation
 474 of 5.5% (Section 2.2).

475 **Shade response.** Fig. 5 shows the simulated relative yield of rainfed
 476 crops under static arrays as a function of the season-mean shading of the crop.
 477 Yield declines close to proportionally with shading: 84–89% relative yield at
 478 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat
 479 loses most and potato and maize least. This response is steeper than the *less*
 480 *than proportional* losses that Laub et al. (2022) report for small shade levels
 481 of C3 cereals and tubers, and it does not reproduce their finding that maize
 482 and grain legumes are more shade sensitive than C3 cereals. In the simulator,
 483 shading also relieves heat and drought in rainfed maize and soybean, which
 484 offsets the higher harvest-index penalty we assigned to them. The absolute
 485 effect of shading on each crop is therefore uncertain (Section 4.4).

486 **Comparison with field trials.** We fitted one radiation-use compen-
 487 sation coefficient per crop to the yield–shading curves of Laub et al. (2022)
 488 using training-block sites only (Methods, calibration script in the repository).
 489 The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the
 490 proportional model already lies within about 0.02 of the C3-cereal and tu-
 491 ber curves, and it cannot lower the maize and legume responses below the
 492 proportional case (root-mean-square error 0.16 for maize and 0.08 for soy-
 493 bean). Against independent field trials (Table 4) the model underestimates
 494 retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and
 495 potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al.
 496 (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91
 497 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was

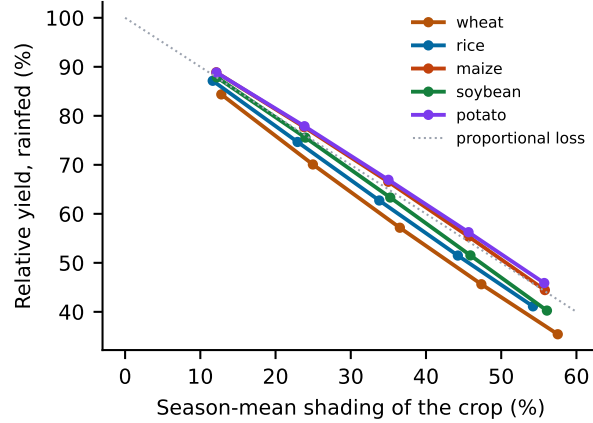


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 5 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains reported in this paper are thus lower bounds with respect to these trials and not upper bounds.

3.7. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 6) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize −31%, rice −27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265),

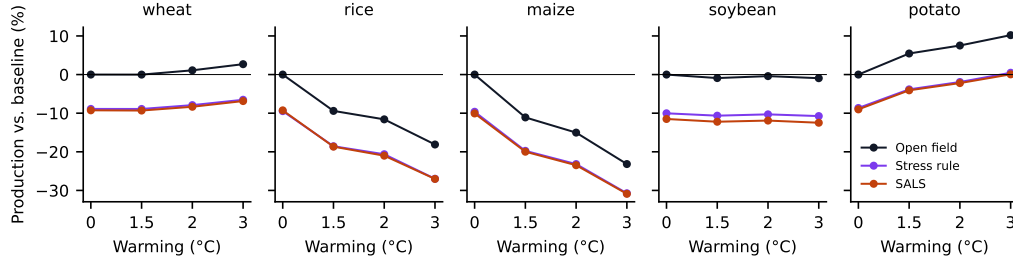


Figure 6: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

518 because the floor allows slightly denser arrays in warmer conditions (GCR
519 0.173 to 0.177 and electricity 510 to 540 MWh ha⁻¹ yr⁻¹). The margin of
520 SALS over the rules does not grow with warming (Table A.6).

521 3.8. Global potential and what it would take

522 Every site is evaluated when it is held out (219 of the 400 crop-site
523 pairs, in at least one split), and each crop’s mean electricity per hectare
524 is scaled to its physical area in SPAM 2010 (581 Mha for the five crops,
525 which avoids counting multiple-cropped land twice). Equipping 5% of this
526 area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr⁻¹
527 under the baseline climate and 16.0 PWh yr⁻¹ at +3 °C, with a production-
528 weighted retention of 0.903 and 0.900. The phenology rule would give 13.0
529 and 14.5 PWh yr⁻¹, and static arrays 6.4 and 6.9 PWh yr⁻¹. For comparison,
530 world electricity generation was about 30 PWh in 2023 (Ember, 2024).

531 These numbers require large amounts of hardware. SALS at 5% cover-
532 age needs 10.6 TWp of modules (11.3 TWp at +3 °C), at a capacity factor of
533 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW
534 (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet.
535 The electricity is also variable, peaks at midday and would be curtailed or
536 stored. The figures are technical potentials, not projections. Ordering sites
537 by electricity per hectare gives a supply curve (Fig. 7): the best 2% of the sta-
538 ple cropland would supply 16 PWh yr⁻¹ and the best 5% 40 PWh yr⁻¹ with
539 SALS. The best tenth of the sites are mostly rainfed maize and soybean sites
540 in the Americas and in southern and eastern Africa (7% mean irrigated frac-
541 tion against 29% overall), which are also among the regions where the floor
542 is hardest to meet (Section 3.5). Siting by electricity alone would therefore
543 concentrate agrivoltaics where food-security risk is highest.

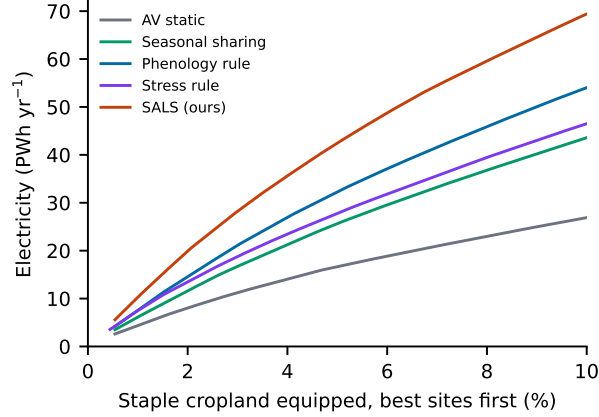


Figure 7: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

544 4. Discussion

545 4.1. What the results establish

546 Three findings stand on consistent evidence. First, sharing light with the
547 crop is the main lever on the electricity that agrivoltaic arrays can produce
548 under a yield floor. With static trackers the floor limits arrays to about 6%
549 ground coverage and an LER of 1.07. Any light-sharing control allows about
550 three times the density, and the LER rises to 1.18–1.24 for hand-written
551 rules. Second, a controller learned through the differentiable simulator is
552 better than the best rule we could write, by 0.03–0.05 in LER and about
553 10% in electricity at equal retention. The gain is small compared with the
554 gap between static and dynamic control. It is, however, reproducible over
555 training seeds (range 0.002 in LER), three spatial splits, three canopy-cooling
556 assumptions and four climates, and it is found at 84–86% of the held-out
557 sites. Third, the learned controller is a smooth function of crop phenology.
558 It shares light almost fully across the middle of the season, and it produces
559 electricity at the beginning and the end. Our ablations show that it does
560 not use stress information, and that it does not need training in warmed
561 climates, to achieve the gain.

562 The modest size of the learned gain should be stated as a result. The
563 phenology rule already contains the main idea, and the learned schedule

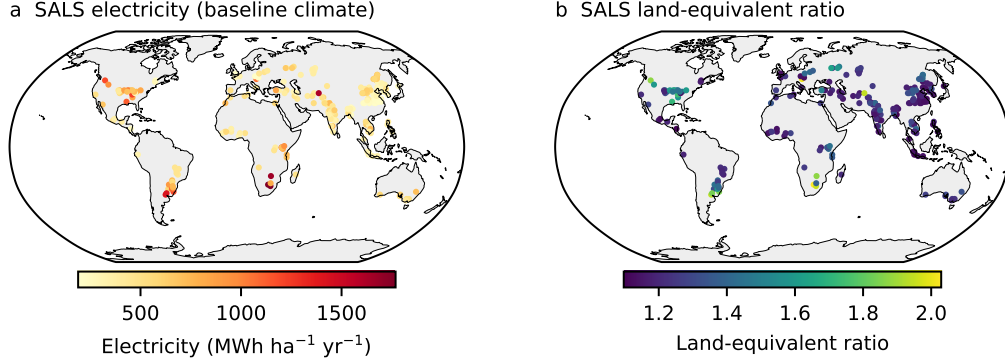


Figure 8: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

refines its timing. A reader who needs a simple, auditable controller can implement the phenology rule and obtain about 85% of the electricity gain of SALS over static arrays. The learned controller has two further properties that we measured (Section 3.3). It was trained once, at a floor of 0.9, and keeps its advantage over the phenology rule at floors between 0.80 and 0.95 (+0.033 to +0.037 in LER under the baseline climate), so one network serves regulations with different floors. Its result is also identical to three decimals for forecast errors up to three times those of training, because it relies on crop phenology, not on weather forecasts, whereas the stress rule degrades with forecast error. These are practical advantages, not larger gains in electricity. Two other routes that we expected to add value did not: conditioning the network on the floor, and chance-constrained training, gave no improvement.

4.2. What the results do not establish

The early hypothesis that stress-aware control, that is, shading the crop on heat and drought days, would be a main source of benefit under warming was not supported. The stress rule performs worse than the phenology rule in every climate, and a learned controller without stress inputs performs as well as one with them. A plausible reason is that in our crop model the yield gain from shading on a hot or dry day is smaller than the loss of light on that day, so stress information does not help. Whether this holds for crop models with more detailed canopy-temperature and heat-stress responses is an open question. Agrivoltaics does not protect staple production against warming in our simulations: light-sharing systems produce about 90% of the

open-field yield in every climate and follow the same decline in maize and rice (Fig. 6).

4.3. Food security in practice

Holding the historical mean retention at 90% does not hold it in each season or at each site. 70% of SALS seasons meet the floor, the lower tail is slightly heavier than for the rules, and production-weighted retention is below 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-constrained design raises season-level compliance to about 92% at a cost of 0.065 in LER, and the ranking of the controllers is preserved. Whichever formulation is chosen, a per-site regulatory floor would need a region- and crop-specific design step. The upscaled potential (Section 3.8) shows the trade-off in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply need about 10 TWp of modules, several times the present global fleet, and the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

4.4. Limitations

- *Simulation only.* All results come from a simulator. PV output was benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop response to shade was only compared with a handful of field trials and one meta-analysis (Section 3.6). The simulated response is close to proportional, is conservative with respect to the trials (rice, wheat, potato, maize) and does not reproduce the stronger losses of maize and legumes in Laub et al. (2022). The harvest-index sensitivities h are assumptions, and the sensitivity analysis uses rule controllers on a subsample of 22 sites.
- *Canopy cooling.* The effect of shade on canopy temperature was represented by one coefficient κ , applied to a heat-stress function that was calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018). The ranking of the controllers is robust to κ between 0 and 2 °C, but the absolute electricity changes by about $\pm 8\%$. Nocturnal warming under modules and rain redistribution are not modelled.
- *Crop model.* SIMPLE has a fixed harvest index and linear radiation-use efficiency. More detailed models with grain-number formation and canopy energy balance may favour different schedules.

- 621 • *Climate scenarios.* Warming was applied as uniform temperature shifts
622 with fixed precipitation, radiation and relative humidity, and with un-
623 changed cultivars and sowing dates. It is not a bias-adjusted climate
624 projection.
- 625 • *Initialisation and state information.* The network is initialised from the
626 phenology rule, because training from random weights collapsed to full
627 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,
628 and we cannot say whether a better optimiser or a richer policy class
629 would find larger gains. The controller uses the simulated state of the
630 crop and soil (thermal time, canopy interception, soil water) without
631 measurement error, and the forecast errors we tested apply to weather
632 only.
- 633 • *Economics and deployment.* There is no cost, market or grid analysis.
634 The value of midday electricity, curtailment, storage, structural cost
635 and farmer acceptance (Trommsdorff et al., 2021) will determine the
636 realisable share. The average row pitch at the chosen densities (about
637 13 m for 2.2-m modules) leaves room for machinery, but the densest
638 designs (GCR up to 0.5) have a pitch of only 4.4 m.

639 5. Conclusions

640 Light sharing, the daily choice between maximum electricity and letting
641 light through to the crop, is the main lever of tracker-based agrivoltaics un-
642 der a food-security floor. In a differentiable simulator of trackers, PV output,
643 crop light, microclimate and crop growth, evaluated fairly on 400 crop-site
644 pairs with the array density of each site chosen from its own history, light-
645 sharing rules allowed about three times the module density of static arrays
646 at a mean yield retention of 90%. They raised the land-equivalent ratio
647 from 1.07 to 1.18–1.24, and a rule that follows crop phenology and needs no
648 weather forecast was the best of them. The simulated shade response is con-
649 servative against the field trials we could use, so these gains are lower bounds,
650 and calibrating the response raised them without changing the ranking of
651 methods. A learned controller added a further 0.03–0.05 LER, reproducibly
652 over seeds, spatial splits, canopy-cooling assumptions and climates, by follow-
653 ing a smooth phenology-dependent schedule and not by reacting to heat or
654 drought. The yield floor is met on average but not in every season or region,
655 and the electricity potential of 5% of staple cropland (about 14 PWh yr⁻¹)

requires about 10 TWp of modules. Future work should validate the crop response to shade under trackers with more field data, notably for maize and legumes where the meta-analysis and the trials disagree, extend the crop model to grain-number formation and canopy energy balance, add costs and grid constraints, and pursue controllers that guarantee the food-security floor per site and per season.

CRedit authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al., 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters (Zhao et al., 2019). The code of the simulator, of SALS and of all experiments, together with the site list, is available from the author’s public repository (<https://github.com/vinhqddang/Artificial-Intelligence-in-Agriculture>).

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Appendix A. Statistical comparison

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: mean of reported seasons; for wheat and potato the two-year range is -18.7 to $+2.7\%$ and -18.2 to $+11\%$.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 5: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table A.6: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92